Tale App: A Social Media Platform for Interactive Storytelling and Multimedia Engagement

Kummari Kavyasri (24885A0543)
Danam Archana (24885A0544)
Angothu Adhisheshu (24885A0547)
Md. Musthabeen (23881A05DR)

May 2025

Submitted under the guidance of A. Sowmya and Pratussha, Assistant Professors
Vardhaman College of Engineering, Hyderabad

Contents

Abstract

Tale is an innovative social media platform designed to promote storytelling and multimedia creativity. It provides features for writing, audio narration, video sharing, meme creation, and interactive learning tools for children. The app includes AI recommendations, user collaboration, and educational content, making it an inclusive space for creators of all ages. The platform combines technology and storytelling in an intuitive and user-friendly interface, and is intended for individuals, educators, and communities.

Acknowledgement

We thank our mentors A. Sowmya and Pratussha for their guidance, and our community partner Mr. K. Rukhmendar for his insights. Gratitude is extended to the Department of Computer Science and Engineering, Principal Dr. J.V.R. Ravindra, and Chairman Dr. Teegala Vijender Reddy for their support. We appreciate the entire team of VCE and our peers who contributed with feedback, encouragement, and collaboration throughout the project development.

1. Introduction

1.1 Motivation

In today's fast-paced digital landscape, traditional social media lacks depth for long-form creative expression. Tale App was created to serve as an all-in-one multimedia storytelling hub. Whether through text, audio, video, or memes, the platform allows individuals to express creativity and collaborate with others.

1.2 Scope

The app is suitable for creators, students, children, and educational institutions. Tale aims to be a platform that encourages participation across demographics and age groups. The scope includes interactive learning, AI-based personalization, and collaborative tools that can be used in classrooms and communities.

1.3 Objectives

- Create a platform for rich multimedia storytelling
- Provide AI tools for personalized content delivery
- Encourage creative collaboration and content sharing
- Ensure child safety and educational usability
- Enable user monetization through in-app economy

1.4 Need Statement

The platform bridges the gap between entertainment and education. Users are demanding more from their digital platforms—a space for creativity, learning, and meaningful interaction.

1.5 Product Realization Process

- Identified problem and defined product scope
- Conducted surveys and brainstorming sessions
- Designed wireframes and interface layouts
- Developed using Flask, HTML/CSS, and SQLite
- Tested and refined using user feedback

2. Product Realization Planning

2.1 Flow Chart

flowchart-placeholder.png

2.2 Development Steps

1. Requirement Analysis and Survey
2. Initial UI Mockup and Feature Prioritization
3. Backend Development and Authentication Setup
4. Frontend Development and API Integration
5. Multimedia Handling and Children's Section
6. Testing, Debugging, and Deployment

2.3 Gantt Chart

gantt-placeholder.png

3. Community Partner Collaboration

3.1 Details

The Tale App was developed with support from the Centre for Community Innovation and Transformation. Mentorship and real-world feedback from Mr. K. Rukhmendar and our faculty advisors played a key role.

3.2 Field Survey Summary

- 60% prefer storytelling with audio/text
- 85% support meme integration
- 90% suggested including a children's section
- 88% support voice features
- 95% emphasized moderation and content safety

3.3 Specifications from Partner

- Simplified navigation and colorful UI
- Voice-based interaction for accessibility
- Content curation tools for teachers
- Secure authentication
- Personalization via AI

4. Design and Development

4.1 Frontend Design

The UI was designed to be clean, responsive, and intuitive. HTML, CSS, Bootstrap, and JavaScript were used for responsive design.

4.2 Backend Design

- Flask used for routing and web server logic
- SQLite for lightweight, easy-to-use database
- Flask-Login for secure sessions and hashed passwords
- File upload capabilities with audio/image filters

4.3 Tools and Technologies

- GitHub – Code collaboration
- VS Code – IDE
- Canva/Figma – Design tools
- Google Forms – User feedback collection
- Google Cloud API – For future AI/voice integration

4.4 Features

- Secure login system with user profile
- Story publishing and discovery
- Meme creation and upload
- Children's section with rhymes, narration, games
- AI recommendations (future scope)

5. Post Realization Activities

5.1 User Feedback Summary

- "Very creative and easy to use" – Student User
- "Add more languages and offline support" – Educator
- "Great for storytelling in class" – Teacher

5.2 Redesign Summary

- Mobile-first redesign for responsiveness
- Added voice-over support and improved kids UI
- Incorporated content reporting system
- Developed wireframes for collaborative storytelling

6. Business Model and Future Work

6.1 Revenue Streams

- Freemium access
- In-app purchases (voice packs, UI themes)
- Subscription (individuals/institutions)
- Ad-free premium model
- Licensing for schools and organizations

6.2 Future Enhancements

- Real-time chat and collaboration
- Voice-to-text story creation
- Offline story downloads
- Leaderboards and gamification for kids

6.3 Paper Submission and Patent

Paper under preparation for submission to IEEE or UGC CARE journals. Patent discussion ongoing for unique content interfaces, voice tools, and AI personalization engine.

7. Conclusion

Tale App was developed to bridge creativity and community. With support from faculty, community partners, and student collaboration, Tale offers a secure, fun, and interactive platform for digital storytelling. Future updates will further enhance interactivity, monetization, and educational adoption.

References

1. Flask: https://flask.palletsprojects.com
2. SQLite: https://www.sqlite.org
3. W3Schools: https://www.w3schools.com
4. GitHub: https://github.com
5. Google Forms: https://docs.google.com/forms
6. Google TTS API: https://cloud.google.com/text-to-speech
7. Firebase: https://firebase.google.com
8. Canva: https://www.canva.com
9. Figma: https://www.figma.com
10. OpenAI API: https://platform.openai.com